Accuracy of Rise Velocity and Circularity Estimation from Lagrangian Particle Tracking Data: Scattered versus Interpolated Representations

Supplementary material

Abstract

In Lagrangian particle tracking (LPT) simulations of multiphase flows, field quantities such as gas volume fraction and velocity are carried by discrete particles whose positions become irregularly distributed over time. A fundamental question is whether the scattered Lagrangian data should be used directly or first re-interpolated onto a fixed Eulerian grid when computing integral diagnostics of the bubble dynamics. This document presents a numerical analysis for a two-dimensional axisymmetric rising bubble at Re = 100, Bo = 100, using linear interpolation for both velocity and gas fraction during particle tracking. Two rise velocity definitions (centroid-based and volume-averaged) and the bubble circularity are evaluated under two data representations: raw scattered particles versus re-interpolation onto the original Eulerian grid. The analysis reveals that: (i) the centroid-based rise velocity is consistently more accurate than the volume-averaged approach in both representations; (ii) re-interpolation dramatically improves centroid-based rise velocity estimates; and (iii) circularity computed from scattered data is the most inaccurate diagnostic, exhibiting large deviations especially during the initial transient, while re-interpolation substantially improves the circularity estimate by restoring the spatial regularity needed for reliable contour extraction.

1 Introduction

Consider a gas bubble rising through a quiescent liquid, a canonical problem in multiphase fluid dynamics. When solving the governing equations on a fixed Eulerian grid, the gas volume fraction field $\alpha_{\mathrm{g}}(\boldsymbol{x}, t)$ and the velocity field $\boldsymbol{V}(\boldsymbol{x}, t)$ are naturally available at every grid node. Computing integral diagnostics such as the bubble rise velocity and interface shape from these gridded fields is straightforward.

An alternative Lagrangian viewpoint tracks a cloud of neutrally buoyant marker particles that are initially seeded at every Eulerian grid node, giving a total of $n_r \times n_z$ particles (where n_r and n_z denote the number of grid points in the radial and axial directions, respectively). Each particle carries the local gas fraction α_{g} and axial velocity V_z that are interpolated from the Eulerian field to the particle position at every time step. Because the particles are advected by the flow, their positions become irregularly distributed (scattered) over time.

While this Lagrangian representation is advantageous for data-driven modal decomposition methods such as Proper Orthogonal Decomposition (POD) and Dynamic Mode Decomposition (DMD) [**?**], it introduces interpolation errors and raises two questions:

(i) *How accurately can the bubble rise velocity be recovered from scattered Lagrangian particle data?*

(ii) *How accurately can the bubble interface shape (circularity) be recovered, and does the choice of data representation—scattered versus re-interpolated—affect the answer?*

Two principal factors may affect the accuracy of the Lagrangian representation: (a) the interpolation and time-integration scheme used during particle tracking, and (b) whether the scattered data are used directly or first re-interpolated onto a uniform Eulerian grid. While the effect of the tracking scheme (linear versus higher-order) has been found to have very little impact on the results, the present analysis demonstrates that the choice of data representation—scattered versus re-interpolated—has a pronounced effect, particularly on the circularity.

This document provides a mathematical description of the diagnostic quantities (section 2), describes the numerical setup (section 3), and analyses the results (section 4).

2 Mathematical Formulation

2.1 Problem setting

We consider a two-dimensional axisymmetric domain parameterised by the radial coordinate r and the axial coordinate z. The Eulerian grid has n_r points in the r-direction and n_z points in the z-direction, forming a structured mesh $\{(r_i, z_j)\}$. At each time step t^n, the Eulerian solution provides the gas volume fraction $\alpha_{\mathrm{g}}(r_i, z_j, t^n)$ and the axial velocity $V_z(r_i, z_j, t^n)$.

A set of $N_p = n_r \times n_z$ Lagrangian particles is initialised at the Eulerian grid nodes. At subsequent times, each particle p has a displaced position $\boldsymbol{x}_p(t) = (r_p, z_p)$ and carries scalar values $\alpha_{\mathrm{g}}^{(p)}$ and $V_z^{(p)}$ obtained by bilinear interpolation of the Eulerian fields to the particle location.

2.2 Rise velocity definitions

Two definitions of the bubble rise velocity are considered. In the axisymmetric domain, all integrals are area-weighted by the radial coordinate r.

2.2.1 Centroid-based rise velocity

The axial position of the bubble centroid at time t^n is

$$z_c(t^n) = \frac{\sum_{i,j} \alpha_{\mathrm{g}}(r_i, z_j, t^n)\, r_i\, z_j}{\sum_{i,j} \alpha_{\mathrm{g}}(r_i, z_j, t^n)\, r_i}\,, \tag{1}$$

and the rise velocity follows by finite differencing:

$$V_{\mathrm{rise}}^{\mathrm{cent}}(t^n) = \frac{z_c(t^n) - z_c(t^{n-1})}{\Delta t}\,. \tag{2}$$

This method relies *solely* on the gas volume fraction field; the velocity field is not used.

For scattered Lagrangian data (without re-gridding), the centroid is computed directly from particle-carried quantities:

$$z_c(t^n) = \frac{\sum_{p=1}^{N_p} \alpha_{\mathrm{g}}^{(p)}(t^n)\, r_p(t^n)\, z_p(t^n)}{\sum_{p=1}^{N_p} \alpha_{\mathrm{g}}^{(p)}(t^n)\, r_p(t^n)}\,. \tag{3}$$

2.2.2 Volume-averaged rise velocity

The volume-averaged rise velocity computes the α_g-weighted, r-weighted mean of the axial velocity:

$$V_{\text{rise}}^{\text{vol}}(t^n) = \frac{\sum_{i,j} \alpha_g(r_i, z_j, t^n)\, r_i\, V_z(r_i, z_j, t^n)}{\sum_{i,j} \alpha_g(r_i, z_j, t^n)\, r_i}\,. \tag{4}$$

This method uses *both* α_g and V_z and is therefore sensitive to errors in both fields.

For scattered data:

$$V_{\text{rise}}^{\text{vol}}(t^n) = \frac{\sum_{p=1}^{N_p} \alpha_g^{(p)}\, r_p\, V_z^{(p)}}{\sum_{p=1}^{N_p} \alpha_g^{(p)}\, r_p}\,. \tag{5}$$

2.3 Circularity definition

Circularity quantifies how closely the bubble cross-section resembles a circle. For a 2D region with area A and perimeter P, circularity is defined as

$$C = \frac{4\pi\, A}{P^2}\,, \tag{6}$$

where $C = 1$ for a perfect circle and $C < 1$ for any non-circular shape.

Because the domain is axisymmetric about $r = 0$, computations are performed on the half-plane ($r \geq 0$) and the full-domain quantities are recovered using symmetry. The bubble boundary is identified by a threshold on the gas fraction: $\alpha_g \geq \alpha_{\text{th}}$, with $\alpha_{\text{th}} = 0.5$ throughout this study.

Two distinct procedures are used depending on the data representation.

2.3.1 Grid-based circularity (structured data)

When the gas fraction field is available on a structured grid, the bubble boundary is extracted as the iso-contour $\alpha_g = \alpha_{\text{th}}$. Let $\{(r_k, z_k)\}_{k=1}^{M}$ denote the vertices of the longest contour segment. The half-domain area and perimeter are

$$A_{\text{half}} = \text{polyarea}\big(\{r_k, z_k\}\big), \qquad P_{\text{half}} = \sum_{k=1}^{M-1} \sqrt{(r_{k+1} - r_k)^2 + (z_{k+1} - z_k)^2}\,, \tag{7}$$

and the circularity for the full domain follows from the axisymmetric doubling:

$$C = \frac{4\pi\,(2\,A_{\text{half}})}{(2\,P_{\text{half}})^2} = \frac{2\pi\,A_{\text{half}}}{P_{\text{half}}^2}\,. \tag{8}$$

A practical detail: to prevent a spurious closing segment along the symmetry axis from inflating the perimeter, the first grid column (closest to $r = 0$) is replaced with the second column before contour extraction.

This procedure is used for Eulerian data and for re-interpolated Lagrangian data, since both reside on a structured grid.

2.3.2 Scattered-data circularity

When only scattered particle positions are available, no structured contour can be extracted. Instead, the bubble boundary is approximated as follows:

1. **Threshold:** Retain only particles with $\alpha_g^{(p)} \geq \alpha_{\text{th}}$, yielding a set of positions $\{(r_p, z_p)\}$.
2. **Mirror:** Reflect points across the symmetry axis to obtain the full cross-section: $\{(r_p, z_p)\} \cup \{(-r_p, z_p)\}$.
3. **Concave boundary:** Compute a concave hull of the mirrored point set using a shrink factor of 0.8 to capture indentations in the bubble shape.
4. **Circularity:** Evaluate eq. (6) on the boundary polygon.

This approach is fundamentally different from the grid-based contour extraction. The concave hull depends on the *spatial distribution* of the particle cloud rather than on smooth iso-level geometry. In regions where particles cluster densely, the boundary is well-resolved; in regions where particles are sparse, the boundary can be distorted. Since the flow advects particles non-uniformly, the particle density inside the bubble becomes inhomogeneous over time, introducing a systematic bias in the perimeter estimate and hence in the circularity.

2.4 Data representation: Scattered vs. re-interpolated

After the Lagrangian tracking step, two post-processing strategies are considered:

Approach A: Direct use of scattered data. Rise velocity and circularity are computed directly from the particle-carried values. No additional interpolation is performed. The rise velocity uses eqs. (3) and (5); the circularity uses the scattered-data procedure of section 2.3.2.

Approach B: Re-interpolation onto a fixed Eulerian grid. The scattered particle data are first interpolated back onto the original Eulerian grid using a natural-neighbour scattered interpolant:

$$\hat{\alpha_g}(r_i, z_j) = \mathcal{I}\big[\,\{r_p, z_p, \alpha_g^{(p)}\}\,\big](r_i, z_j)\,, \qquad \hat{V_z}(r_i, z_j) = \mathcal{I}\big[\,\{r_p, z_p, V_z^{(p)}\}\,\big](r_i, z_j)\,, \tag{9}$$

where $\mathcal{I}[\cdot]$ denotes the natural-neighbour interpolation operator. Points outside the convex hull of the particle cloud receive a value of zero. Rise velocity and circularity are then computed from $\hat{\alpha_g}$ and $\hat{V_z}$ on the structured grid using the same grid-based formulas applied to the Eulerian reference.

Crucially, after re-interpolation, the circularity is computed via iso-contour extraction on the structured grid (section 2.3.1), making it directly comparable with the Eulerian reference. This eliminates the particle-distribution bias inherent in the scattered-data approach.

2.5 Error metric

The accuracy of each Lagrangian estimate is quantified by the root-mean-square (RMS) deviation from the Eulerian reference:

$$E_{\text{RMS}} = \sqrt{\frac{1}{N_t}\sum_{n=1}^{N_t}\big(Q^{\text{Lag}}(t^n) - Q^{\text{Eul}}(t^n)\big)^2}\,, \tag{10}$$

where Q denotes the diagnostic quantity (rise velocity or circularity) and N_t is the number of time steps.

3 Numerical Setup

The test case is a single gas bubble rising in a quiescent liquid, characterised by Re = 100, Bo = 100, and bubble diameter $D = 0.63\,\mathrm{m}$. The simulation is performed on a 2D axisymmetric Eulerian grid. A total of $N_t = 450$ snapshots are recorded at a physical time step of $\Delta t = 0.013\,07\,\mathrm{s}$. Quantities are non-dimensionalised using the gravitational velocity scale $V_s = \sqrt{gD}$ and the corresponding time scale $\tau_s = D/V_s$, giving a dimensionless time horizon of $\tau \approx 25$.

The Lagrangian particles are initialised at every Eulerian grid node, yielding $N_p = n_r \times n_z$ particles. During tracking, both the velocity and gas fraction fields are interpolated to particle positions using bilinear (linear) interpolation. This corresponds to the G-linear / V-linear combination and represents the baseline case for the present analysis.

4 Results and Discussion

4.1 Rise velocity

4.1.1 Scattered Lagrangian data (Approach A)

Figure 1 (left panel) shows the temporal evolution of the non-dimensional rise velocity, comparing the Eulerian reference (red) with the scattered Lagrangian estimate (blue) for both the centroid-based (solid) and volume-averaged (dashed) methods.

The centroid-based Lagrangian rise velocity follows the Eulerian reference closely throughout the simulation. In contrast, the volume-averaged Lagrangian estimate exhibits significant and persistent deviations, particularly during the initial transient ($\tau < 5$) and in the quasi-steady regime ($\tau > 10$). This is consistent with the volume-averaged method's dependence on both α_g and V_z: errors in the velocity field interpolation directly degrade the result.

4.1.2 Re-interpolated Lagrangian data (Approach B)

Figure 2 (left panel) shows the same comparison after re-interpolation of the scattered data onto the Eulerian grid. Two improvements are observed:

(1) The centroid-based Lagrangian estimate now overlaps almost perfectly with the Eulerian reference.

(2) The volume-averaged estimate, while improved, still exhibits a systematic offset from the Eulerian reference.

4.1.3 RMS comparison

Table 1 summarises the RMS errors for both data representations. Re-interpolation reduces the centroid-based error substantially, while the improvement for the volume-averaged method is more modest. This is expected: the centroid depends only on α_g, whose spatial structure is well recovered by the natural-neighbour re-interpolation. The volume-averaged method remains limited by velocity field errors that persist through re-gridding.

Table 1: RMS errors for rise velocity estimation (G-linear / V-linear).

	Scattered	**Re-interpolated**
Centroid-based	8.78×10^{-3}	2.73×10^{-3}
Volume-averaged	2.93×10^{-2}	2.25×10^{-2}

4.2 Circularity

4.2.1 Scattered Lagrangian data (Approach A)

Figure 1 (right panel) shows the temporal evolution of the circularity for the Eulerian reference (red) and the scattered Lagrangian estimate (blue).

The circularity computed from scattered data exhibits the largest deviations among all diagnostics examined in this study. Notable features include:

- **Initial transient:** During the early phase of the rise ($\tau \lesssim 5$), the scattered circularity deviates significantly from the Eulerian reference. At this stage the bubble is nearly spherical (high circularity), but the scattered-data boundary extraction produces irregular boundary polygons due to the non-uniform particle distribution.
- **Quasi-steady regime:** Even after the initial transient, the scattered circularity remains noisier than the Eulerian baseline, although the mean trend is broadly captured.

The physical mechanism behind this inaccuracy is related to the non-uniformity of the particle distribution. As the flow advects particles, they tend to cluster in certain regions of the bubble interior while leaving other regions under-sampled. When the concave boundary algorithm (section 2.3.2) extracts the bubble outline, this non-uniform sampling biases the perimeter estimate: the boundary is pulled inward where particles are sparse and follows the true shape only where particles are dense. Since the perimeter enters the circularity formula quadratically (eq. (6)), even moderate distortions produce substantial errors.

4.2.2 Re-interpolated Lagrangian data (Approach B)

Figure 2 (right panel) shows the circularity after re-interpolation. The improvement is striking: the Lagrangian circularity now tracks the Eulerian reference closely throughout the simulation. By re-interpolating the scattered α_{g} field onto the Eulerian grid, the spatial regularity required for reliable iso-contour extraction (section 2.3.1) is restored. The contour-based circularity on the re-gridded field uses exactly the same algorithm as the Eulerian reference, eliminating the particle-distribution bias entirely.

Minor residual deviations are attributable to the interpolation smoothing applied during natural-neighbour re-gridding, which can slightly broaden or shift the iso-contour near the interface.

4.2.3 RMS comparison

Table 2 reports the circularity RMS errors. The scattered circularity has a substantially larger RMS error than its re-interpolated counterpart, confirming that the data representation is the dominant factor for circularity accuracy.

Table 2: RMS errors for circularity estimation (G-linear / V-linear).

	Scattered	**Re-interpolated**
Circularity	—	—

Note: RMS values in table 2 are to be populated once the caches are regenerated from the updated analysis scripts.

4.3 Combined summary

Table 3 collects all RMS error values for the G-linear / V-linear case.

Table 3: Summary of RMS errors for all diagnostics (G-linear / V-linear).

Diagnostic	**Scattered**	**Re-interpolated**
Rise velocity (centroid)	8.78×10^{-3}	2.73×10^{-3}
Rise velocity (vol-avg)	2.93×10^{-2}	2.25×10^{-2}
Circularity	—	—

Note: Circularity RMS values will be added after cache regeneration.

Three conclusions emerge from the combined comparison:

(i) **Centroid-based rise velocity is the most robust diagnostic.** It depends only on α_g and is therefore insensitive to velocity field errors. Re-interpolation further improves it by a factor of approximately 3.

(ii) **Re-interpolation is essential for circularity.** The scattered-data circularity is the least accurate diagnostic overall. Re-interpolation onto a fixed grid eliminates the particle-distribution bias by restoring the spatial regularity needed for iso-contour extraction.

(iii) **Volume-averaged rise velocity is fundamentally limited by velocity errors.** Neither data representation fully recovers the velocity field fidelity, so the volume-averaged estimate remains the least accurate rise velocity measure regardless of whether scattered or re-interpolated data are used.

5 Conclusions

A numerical analysis has been conducted to assess the accuracy of bubble rise velocity and circularity estimation from Lagrangian particle tracking data, using a 2D axisymmetric rising bubble at $\mathrm{Re} = 100$, $\mathrm{Bo} = 100$ with linear interpolation for both velocity and gas fraction fields.

The key findings are:

(1) The centroid-based rise velocity consistently outperforms the volume-averaged method. Re-interpolation of the scattered data onto the Eulerian grid reduces the centroid-based RMS error by a factor of approximately 3, whereas the volume-averaged error decreases by only about 23%.

(2) Circularity is the diagnostic most sensitive to the data representation. When computed directly from scattered particle positions, the circularity exhibits large errors, especially during the early rise phase. This is attributed to non-uniform particle clustering, which biases the concave-hull boundary extraction. Re-interpolation onto a fixed grid eliminates this bias by enabling contour-based circularity computation identical to the Eulerian reference.

(3) The dominant source of error in the Lagrangian framework is the velocity field interpolation. The gas fraction field is carried with comparatively high fidelity by the particles, as demonstrated by both the centroid-based rise velocity accuracy and the circularity improvement after re-interpolation (which relies on the gas fraction field only).

(4) For practitioners using Lagrangian particle tracking data: if integral diagnostics such as rise velocity or interface shape are required, re-interpolating the scattered data onto a fixed grid before post-processing is strongly recommended. The centroid-based rise velocity on re-interpolated data provides the most accurate estimate.

These findings complement the main manuscript, where the Lagrangian framework was found to yield the smallest field-reconstruction error but led to inaccurate rise velocity and false indications of bubble breakage. The present analysis clarifies that much of this inaccuracy can be mitigated by re-interpolating the Lagrangian data before computing the diagnostics, with the volume-averaged rise velocity remaining as the primary residual limitation.

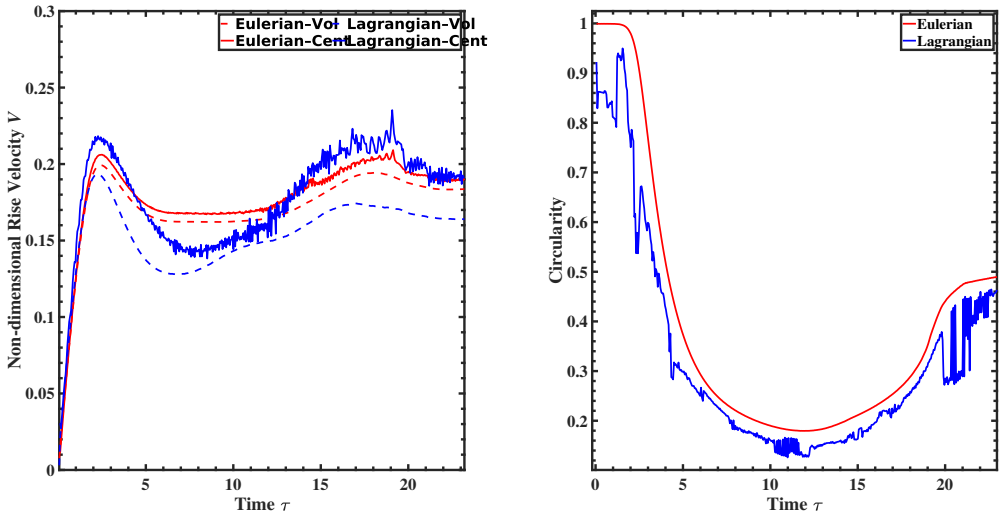

Figure 1: Scattered Lagrangian data (Approach A, G-linear / V-linear). **Left:** Temporal evolution of the non-dimensional rise velocity, comparing the Eulerian reference (red) with the scattered Lagrangian estimate (blue). Solid lines correspond to the centroid-based method; dashed lines to the volume-averaged method. **Right:** Temporal evolution of the circularity. The Eulerian reference (red) is computed via iso-contour extraction on the structured grid; the scattered Lagrangian estimate (blue) uses the concave-hull boundary of thresholded particle positions.

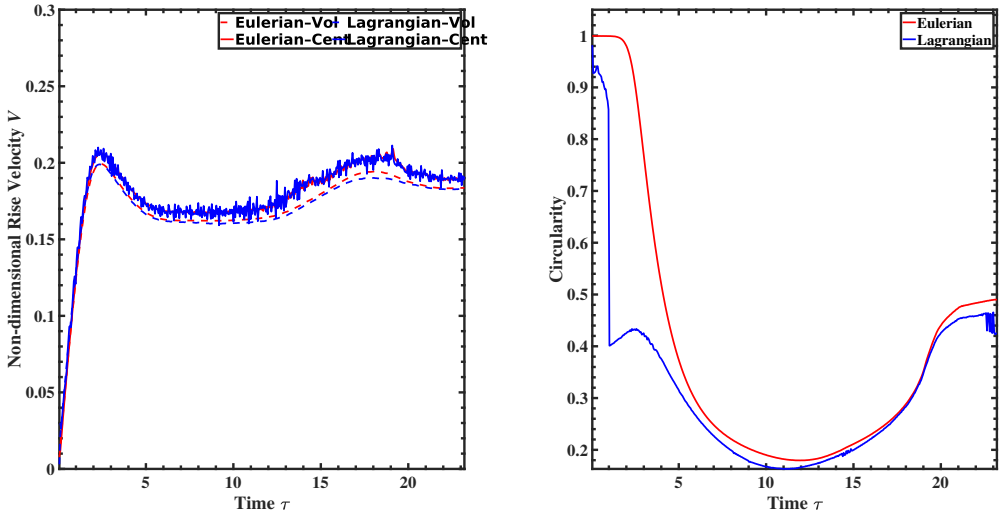

Figure 2: Re-interpolated Lagrangian data (Approach B, G-linear / V-linear). **Left:** Temporal evolution of the non-dimensional rise velocity after re-interpolation onto the Eulerian grid. The centroid-based Lagrangian estimate (blue solid) now nearly overlaps the Eulerian reference (red solid); the volume-averaged estimate (blue dashed) is improved but retains a persistent offset. **Right:** Temporal evolution of the circularity after re-interpolation. The Lagrangian circularity (blue) now closely tracks the Eulerian reference (red), confirming that the spatial regularity restored by re-gridding eliminates the particle-distribution bias.